DP Biology – B2.3 (HL extension)

Specialized cells

- What do stem cells do in multicellular organisms?
- How do differentiated cells suit their specialized roles?

HL statements B2.3.7–10. Textbook pp. 265–272.

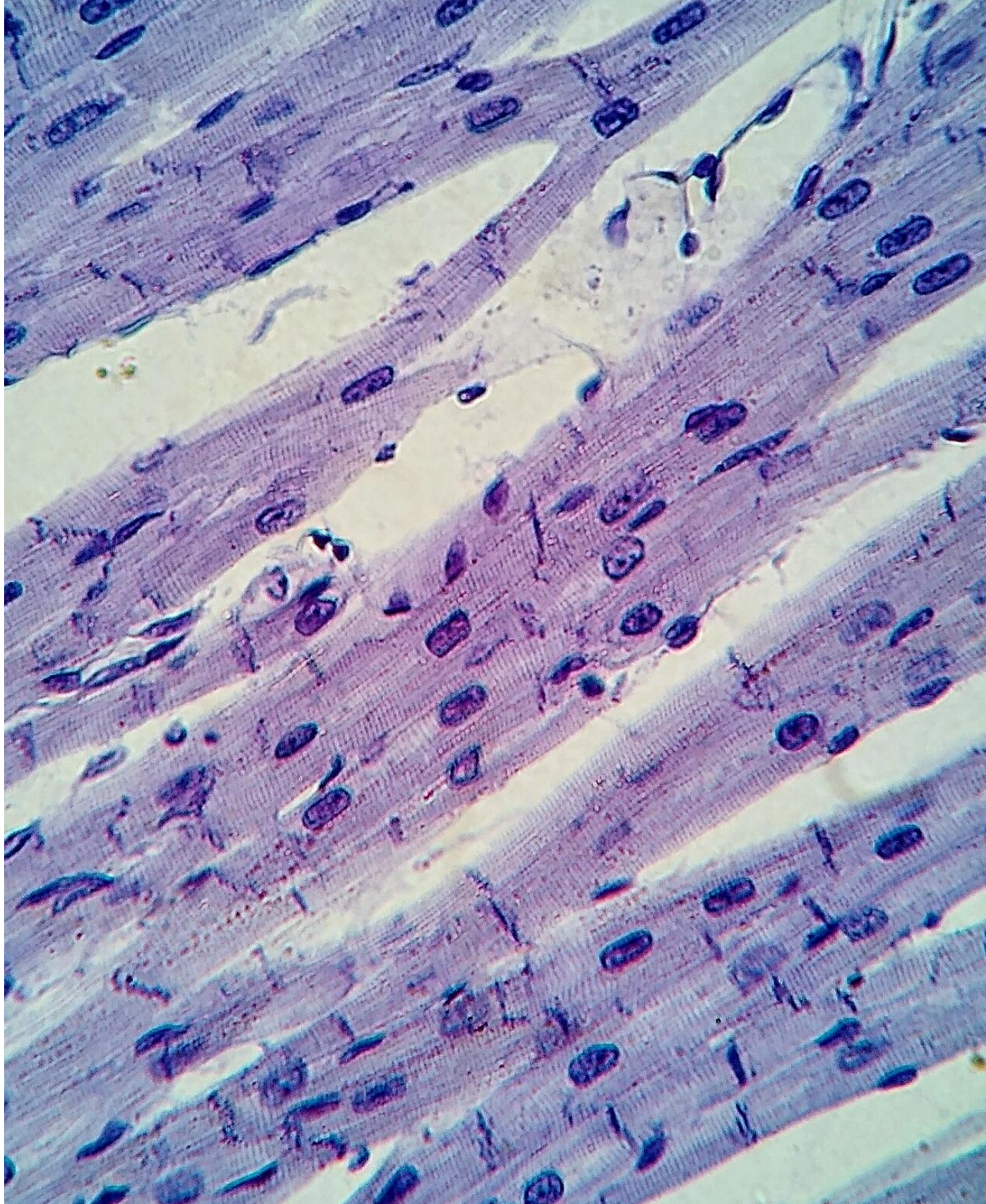


Image: heart muscle cells (light micrograph). A. G. Cheroske, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B2.3.7–8	Explain how cells increase their surface area-to-volume ratio, and how the two kinds of pneumocyte suit their roles
2	B2.3.9–10	Compare heart muscle cells with skeletal muscle fibres, and explain the adaptations of human sperm and egg cells

The big picture: shape follows job

Exchange	Contraction	Reproduction
Red cells, kidney tubule, pneumocytes	Heart and skeletal muscle	Sperm and egg

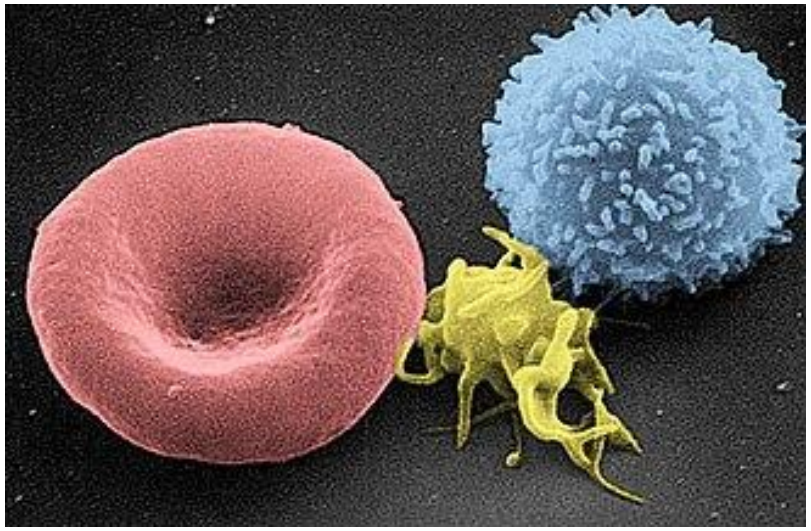
Each specialized cell has **structures** that fit its **function**. Look for surface area, organelles, size and shape.

Big question

Why is an egg about 20 times longer than a sperm head, and made in far smaller numbers?

More surface for the same volume

B2.3.7



Red blood cell (left) and white blood cells (SEM). NCI, public domain.

Adaptation	How it helps
Flattening	More surface per volume; short distances
Microvilli	Finger-like projections: brush border
Invagination	Infolding, often for active transport

Red blood cells

Biconcave and without a nucleus: oxygen diffuses a short way to every haemoglobin, and the cell bends through capillaries.

Same volume, more surface

B2.3.7



cube: SA:V low



flattened: SA:V higher

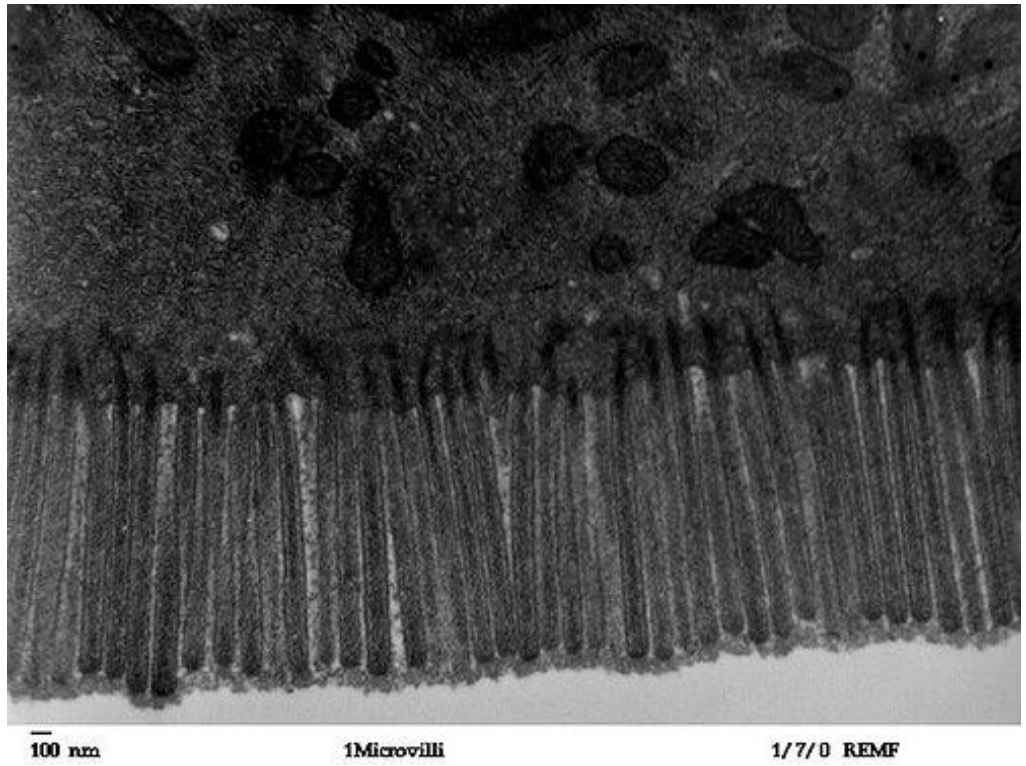


microvilli: higher still

So

Flattening and folding add membrane without adding volume, speeding diffusion and transport.

Proximal convoluted tubule cells



Microvilli forming a brush border (TEM). L. Howard and K. Connollly, public domain.

Microvilli face the filtrate: area for cotransporters. **Invaginations** face the blood: area for pumps.

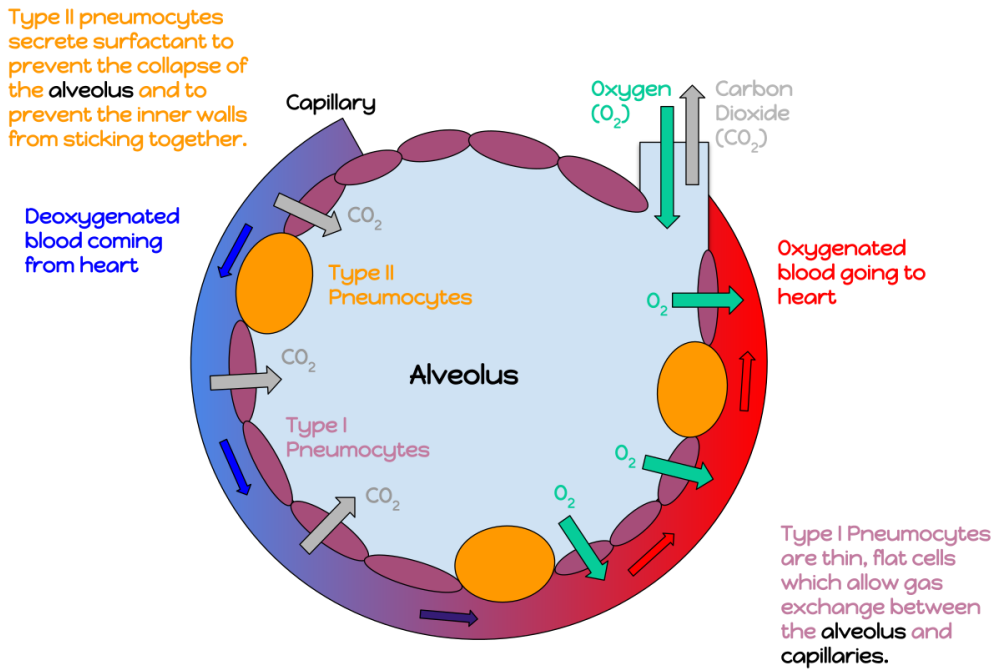
Many **mitochondria** supply ATP for active transport.

Link

Same features in gut lining; Na/K pumps and cotransporters (B2.1).

Two cell types line each alveolus

B2.3.8



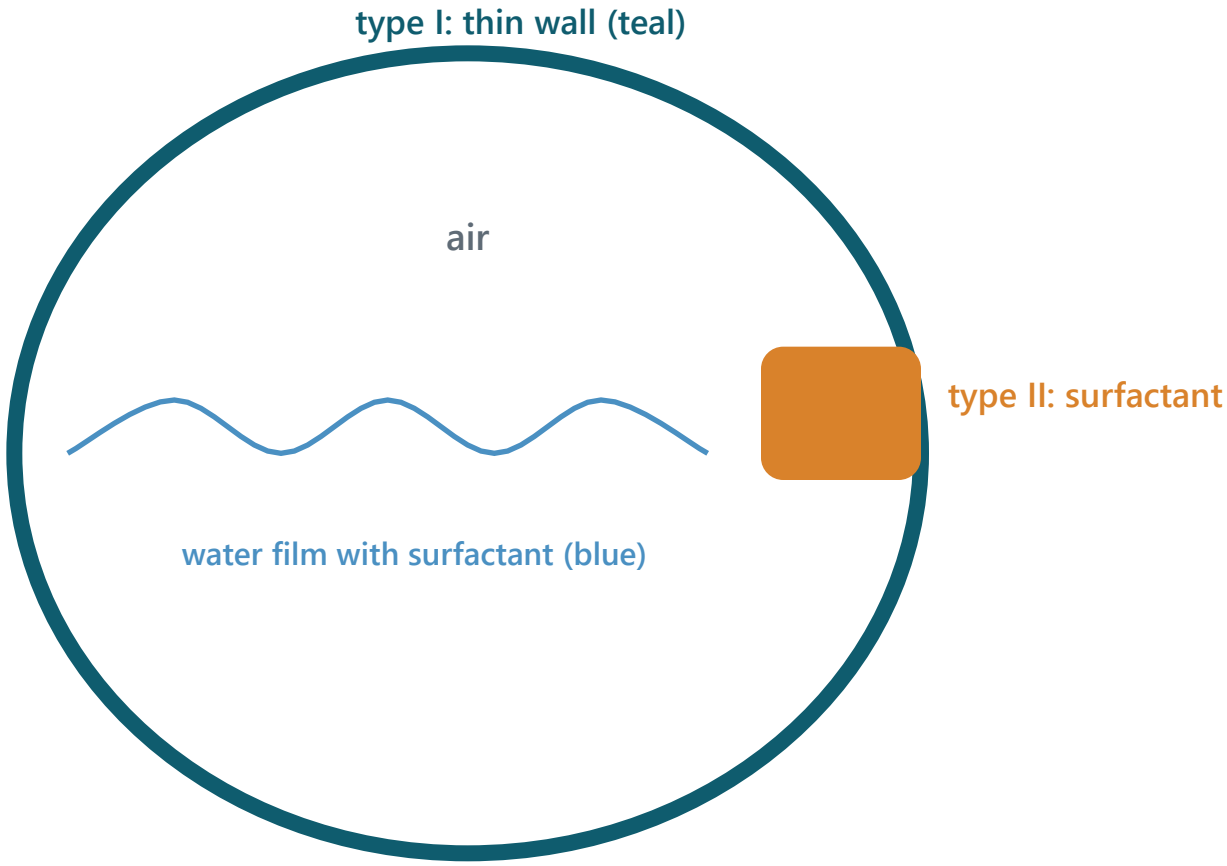
	Type I	Type II
Shape	Very thin, flat	Cuboidal
Surface share	About 95%	About 5%
Features	Few organelles	Lamellar bodies, RER, Golgi
Role	Gas exchange	Secrete surfactant

Alveolus and its cell types. Katherinebutler1331, CC BY-SA 4.0.

Two jobs

Very thin walls and a supply of surfactant: no single cell type could do both well.

Inside an alveolus



Surfactant

Spreads over the water film and lowers surface tension, so alveoli do not collapse or stick when we breathe out.

Lamellar bodies

Store surfactant until exocytosis.

Surfactant and premature babies

B2.3.8

Case study

Very premature babies may lack enough type II pneumocytes making surfactant. Alveoli collapse and breathing is hard (respiratory distress).

Treatment

Artificial surfactant into the lungs; steroids given to the mother speed surfactant production.

Think

Why do type II cells have more mitochondria and rough ER than type I cells?

Working with data: surfactant and premature babies

B2.3.8

Babies with breathing difficulties per 100 births, by weeks of pregnancy (rounded, typical values).

Weeks at birth	Per 100 births
26	90
30	50
34	10
39	1

1. Describe the relationship.
2. Explain it using type II pneumocytes.
3. Suggest why steroids are given before an early birth.

Check-in: class 1

B2.3.7–8

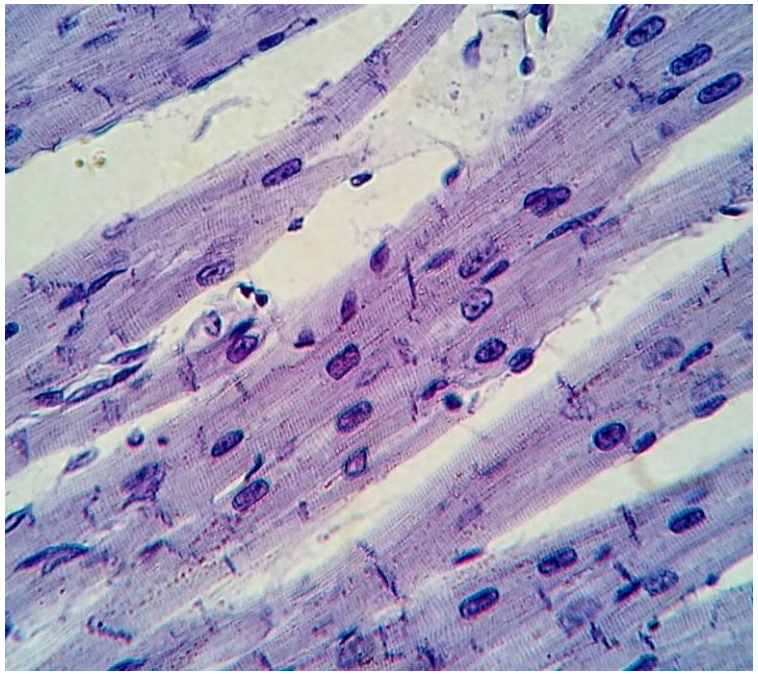
1. Give three ways cells raise SA:V.
2. Explain the adaptations of PCT cells.
3. Compare the two kinds of pneumocyte.
4. Explain the role of surfactant.
5. Explain why premature babies may struggle to breathe.

How to use this

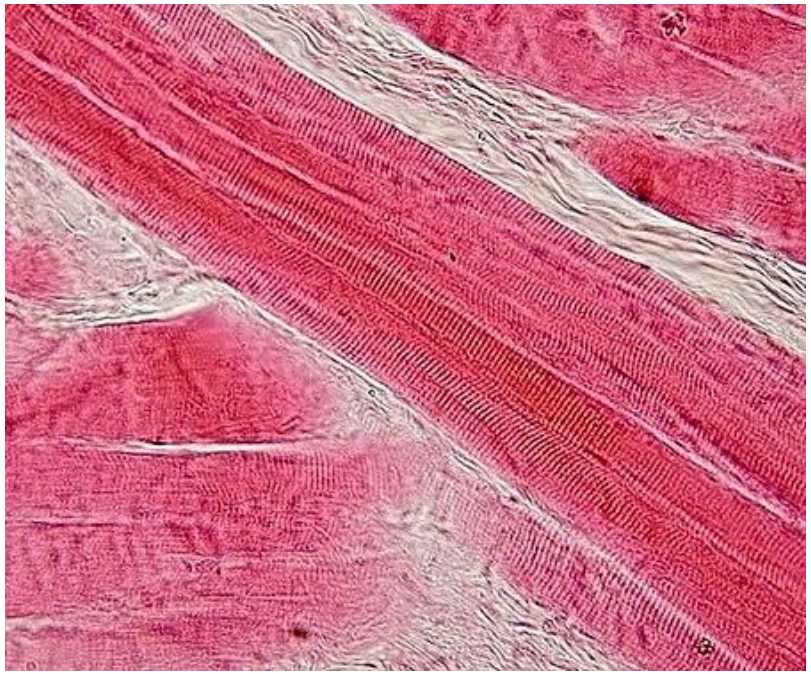
Answer without your notes, then check against slides 3–10.

Heart muscle and skeletal muscle

B2.3.9



Cardiac muscle (light micrograph). A. G. Cheroske, CC BY-SA 4.0.



Striated (skeletal) muscle. CC BY-SA 3.0.

Shared

Both striated (sarcomeres of actin and myosin) with many mitochondria.

Look

Find the branches and intercalated discs in heart muscle, and the many nuclei in skeletal fibres.

Branched cells and long fibres

B2.3.9



cardiac: branched; disc (black); one nucleus



skeletal fibre: long, unbranched, many nuclei

Hypotheses

Branching spreads the signal so the heart contracts together. Long fibres shorten far along one line of pull. Many nuclei supply a huge cytoplasm.

Heart and skeletal muscle compared

B2.3.9

	Cardiac muscle cell	Striated muscle fibre
Shape	Short and branched	Long and unbranched
Nuclei	One (or two)	Many: cells fused
Junctions	Intercalated discs, gap junctions	None between fibres
Control	Contracts on its own	Needs a motor neuron

Exam tip

Give paired differences: state both sides.

Is a muscle fibre a cell?

B2.3.9

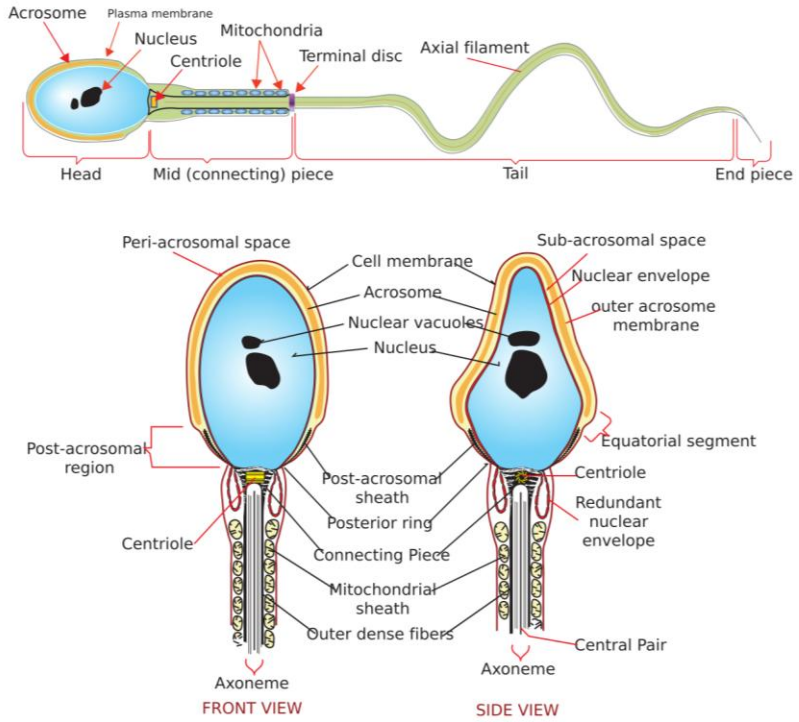
Like a cell	Not a typical cell
One continuous plasma membrane	Many nuclei
Usual organelles	Formed by fusion of many cells
Works as one unit	Far larger than most cells

Discuss

Does a muscle fibre break cell theory, or show that the theory needs careful wording? Give your best evidence.

The human sperm cell

B2.3.10



Human sperm cell. Mariana Ruiz Villarreal, public domain.

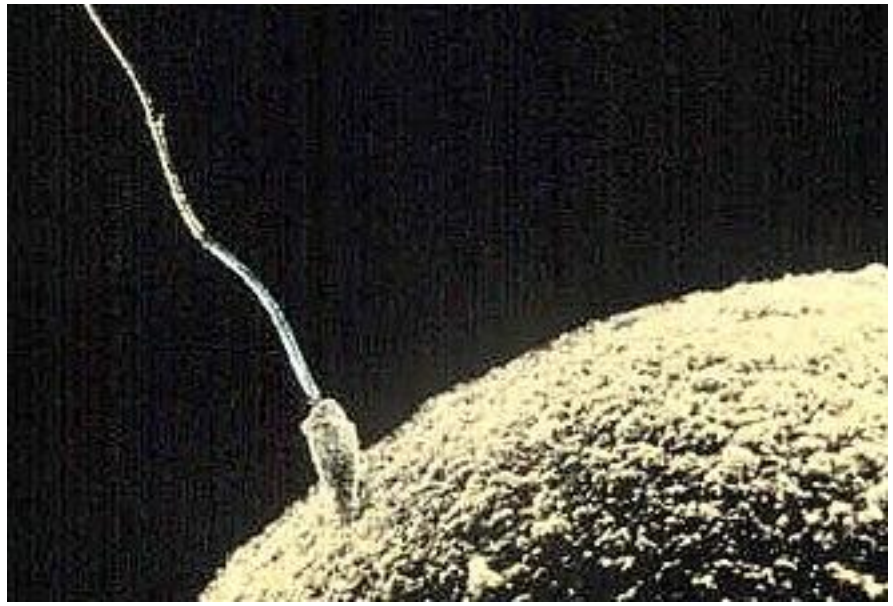
Structure	Adaptation
Acrosome	Enzymes digest the zona pellucida
Haploid nucleus	Packed DNA: small head
Mid-piece	Mitochondria: ATP to swim
Flagellum	9 + 2 microtubules
Little cytoplasm	Streamlined; made in millions

Mitochondria

Sperm mitochondria are destroyed after fertilization, so mtDNA comes from the mother.

The human egg cell

B2.3.10



Sperm meeting an egg (SEM). Public domain.

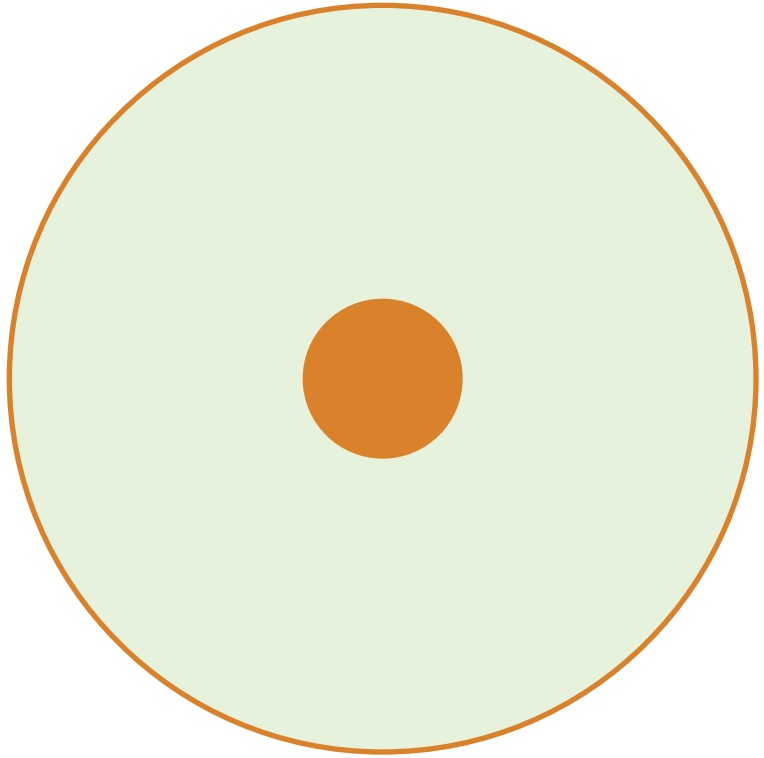
Structure	Adaptation
Large cytoplasm	Nutrients, organelles, mRNA
Haploid nucleus	Arrested in metaphase II
Zona pellucida	Binds sperm; hardens to block more
Cortical granules	Harden the zona after one sperm
Follicle cells	Protect and feed

Polyspermy

Cortical granules ensure only one sperm fertilizes the egg.

Two gametes, two strategies

B2.3.10



egg: about 100 µm



sperm head about 5 µm

	Egg	Sperm
Size	Large	Tiny
Number	Few	Millions
Moves?	No	Swims

Big question

Eggs stock the embryo; sperm compete to arrive. Different jobs, different designs.

Practice: muscle and gametes

B2.3.9–10

Compare

Cardiac and skeletal muscle: three pairs.

Explain

Two sperm adaptations.

Explain

How the egg blocks a second sperm.

Use

intercalated disc, multinucleate, acrosome, cortical granules

Check-in: class 2

B2.3.9–10

1. Give three differences between cardiac and skeletal muscle.
2. Explain whether a muscle fibre is a typical cell.
3. State the role of the acrosome.
4. State the role of cortical granules.
5. Explain why sperm have many mitochondria.

How to use this

Answer without your notes, then check against slides 12–19.

Common mistakes to avoid

Not this	But this
Type II pneumocytes do gas exchange.	Type I do; type II make surfactant.
Cardiac muscle is not striated.	Both are striated.
Sperm and egg are similar in size.	The egg is far larger.

Summary: specialized cells

B2.3.7–10

	Key idea
B2.3.7	SA:V raised by flattening, microvilli, invaginations
B2.3.8	Type I thin; type II surfactant
B2.3.9	Cardiac branched, discs; skeletal long, multinucleate
B2.3.10	Sperm: acrosome, mid-piece; egg: zona, cortical granules

Check that you can...

1. **Explain** how cells raise SA:V
2. **Explain** pneumocyte adaptations
3. **Compare** cardiac and skeletal muscle
4. **Explain** gamete adaptations

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State two features of type I pneumocytes that suit gas exchange.
2. Give two differences between heart muscle cells and skeletal muscle fibres.
3. State the role of the acrosome and of cortical granules.

Image credits

Cardiac muscle histology 400x.jpg – Alexander G. Cheroske – CC BY-SA 4.0

Red White Blood cells.jpg – Electron Microscopy Facility at The National Cancer Institut – Public domain

Human jejunum microvilli 1 – TEM.jpg – Louisa Howard, Katherine Connollly – Public domain

An annotated diagram of an alveolus.svg – Katherinebutler1331 – CC BY-SA 4.0

Skeletal muscle – longitudinal section.jpg – Author not stated – CC BY-SA 3.0

Complete diagram of a human spermatozoa en.svg – Mariana Ruiz Villarreal – Public domain

Sperm-egg.jpg – Unknown author – Public domain

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